

Sreeram_experimental — Branch Overview

BUSI70575 (Imperial) · meta-labelling metamodel · 11 instruments (equity / energy / metals) · 4886 events · 2026-06-03

Purpose: assessment-of-record for which model variant to submit. Not the final academic report — concise overview of what was built, what was measured, and the decisions made.

1. Data

Source	What	Range	Use
data/ohlcv_data.csv	Daily OHLCV for 11 futures	1990-01-02 → 2022-06-30 (8200 rows)	Labels: F1-F17 + F21 features
data/primary_signals.csv	Primary signal in {-1, 0, +1}	2020-01-03 → 2022-06-30 (645 dates)	Label trigger; trade direction
data/bloomberg/cleaned/macro_harry.parquet	Harry's 21 macro series (VIX, rates, credit, FX, EIA, COMEX)	1990-01-02 → 2022-06-30 daily	F11 macro features (rank/changes)
data/bloomberg/cleaned/futures_term.parquet	BBG raw front (CL1..PL1) + 2nd-month (CL2..PL2) + 10-year (CL10..PL10); XB from F12025	1990-01-02 → 2022-06-30	F12 term structure (NEW)
data/bloomberg/cleaned/options_iv.parquet	1m/3m ATM IV + 90%/110% MNY for 10 underly	2005-2006 → 2022-06-30	F19 options IV (NEW)
data/bloomberg/cleaned/eia_crude.parquet	EIA weekly crude inventory CHANGES (DOEASCRD)	1990-01-02 → 2022 (Friday data, +5d release)	F22 EIA flag + crude_change_z26w (NEW)

Hidden test: H2-2022 — grader re-runs the pipeline; BBG data CANNOT be extended past 2022-06-30 (released period, plan \$5.1). R-11 ablation ensures model robustness to BBG missingness at hidden-test time.

R-10: OHLCV is **ratio back-adjusted continuous futures**, not raw front-month. Return-R² vs raw BBG over 2020-2022: metals 0.995+, equity 0.94-0.97, ng1s 0.72 (problematic — 11% label flip).

2. Features (105 registered, 80 kept after drift filter)

Family	N reg.	N kept	Source / lift	Notes
F1 counter-trend	3	3	Sreeram + shared	bb_pctb, rsi_14, mr_score
F2 vol / dispersion	6	6	Sreeram + alken	vol_20/60, vol_ratio, parkinson, GK, vol_of_vol
F5 signal trajectory	6	6	Harry signal_trajectory.py	run_length, days_since_flip, entropy, flip_rate, long_bias, participation
F6 momentum	6	5	Sreeram	ts_mom 20/60, MA cross, MACD + signal + hist
F7 microstructure	7	7	Harry microstructure_fixed.py	volume_z, volume_trend, oi_level, oi_change, amihud, kyles_lambda, overni
F8 calendar	4	4	Sreeram	dow_sin/cos, month_sin/cos
F9 cross-section	4	4	Harry cross_asset.py	xsect_rank, dispersion_z, pair_corr_mean, implied_corr_z
F10 price action	4	4	Sreeram	hl_range, oc_ret + 20d means
F11 macro (REFORMULATED)	35	17	Rebuilt — 63d rolling RANKS + chg5 / chg20 levels → ranks per plan \$3.3 (fix for catastrophic train→test drift in a	
F12 path structure	5	5	Sreeram	variance_ratio_5_21, autocorr_21, trend_tval_21, efficiency_ratio_21, hurst_
F15 conditional risk	2	2	Harry conditional_risk.py	path_tortuosity_20, semi_vol_ratio_20
F16 concept drift	1	1	Harry concept_drift.py	regime_alignment_score (proxy version)
F17 HMM (Sreeram)	3	0	Sreeram regimes.py	All-NaN in production (hmmlearn convergence)
F18 term structure (NEW)	5	4	Bloomberg Block A	term_spread, z63, contango_flag, term_chg5, roll_yield
F19 options IV (NEW)	6	6	Bloomberg Block B	atm_iv_1m/3m, term_slope, iv_rv_spread, skew, pctile_252
F21 cross-asset RV	4	2	Plan new (OHLCV-derived)	gold/silver z, copper/gold, crack 3-2-1
F22 event flag (PARTIAL)	2	1	Bloomberg Block E	EIA crude release flag (Wed); change z26w drift-dropped
EWMA HMM (online)	2	2	alken regime.py	Causal online filter, no CV seam
TOTAL	105	80	—	Drift filter: drop if KS>0.25 ∧ val_AUC<0.54 OR KS>0.20 ∧ val_AUC<0.51

Top 8 features by val-AUC (post-drift): f12_variance_ratio_5_21 (0.590), f12_autocorr_21 (0.584), f5_long_bias_20 (0.570), f15_path_tortuosity_20 (0.567), f5_trailing_run_length (0.565), f5_days_since_flip (0.565), **f19_iv_pctile_252 (0.559 — NEW BBG)**, f5_signal_entropy_20 (0.558).

R-11 BBG missingness ablation result: with_bbg vs without_bbg deltas all ≤ 0.011 across classes — model is BBG-robust. All 3 asset classes ship with_bbg.

3. Labels (triple-barrier, t+1 entry)

Spec: pt = sl = 0.5 (symmetric tight); h = 10 trading days; entry at t+1 (Harry's load-bearing off-by-one fix); GARCH(1,1) one-step-ahead daily sigma (refit every 21d, min 500 obs, max window 2000); per-instrument concurrency for AFML Ch.4 uniqueness weights; events with truncated window dropped.

Count: 4886 events exact match to Harry's events.csv (per-instrument also exact).

Instrument	n_events	n_long	n_short	pos_rate	PT %	SL %	Vert %	mean uniq
cl1s	411	375	36	0.672	55.9	29.0	15.1	0.214
es1s	564	446	118	0.530	47.7	41.3	11.0	0.202
fesx1s	626	286	340	0.534	47.0	37.9	15.2	0.178
gc1s	161	129	32	0.602	52.2	33.5	14.3	0.352
hg1s	617	298	319	0.509	45.5	44.2	10.2	0.198
ho1s	63	53	10	0.635	58.7	34.9	6.3	0.466
ng1s	120	0	120	0.517	45.0	40.0	15.0	0.274
nq1s	593	397	196	0.567	51.8	40.3	7.9	0.203
pl1s	547	411	136	0.554	48.6	38.9	12.4	0.213
rb1s	617	358	259	0.525	45.7	41.5	12.8	0.184
si1s	567	297	270	0.510	40.7	43.2	16.0	0.190
TOTAL	4886	3050	1836	0.547	47.8	39.7	12.5	—

Ablation NOT shipped: Harry's wider asymmetric spec (pt=1.5, sl=1.0, h-day cumulative GARCH) produced 69.5% vertical, lifted ng1s to 0.800 AUC but per R-10 that's construction-artefact-driven. Artefacts at `data/sreeram_experimental_events_harry_spec.parquet` for inspection.

4. Models

Family	Used as	Config	Notes
Elastic-net Logistic	Linear baseline (rubric)	saga solver, l1_ratio=0.5, C=1.0; standardised + median-imputed	Individual class champions (ho1s, ng1s, si1s)
XGBoost	Tree baseline (rubric, Family A)	PS5 cell-43: max_depth=4, lr=0.05, subsample=0.8, colsample_bytree=0.8	Colsample (0.8) to pick features; XGB was best CV (1.5%)
LightGBM	Tree (alken parity)	num_leaves=15, lr=0.05, subsample=0.8, deterministic=True	Controls @ individual pool
RandomForest	Tree-bagging	n_estimators=200, max_depth=6, min_samples_leaf=10	100+ sorted features (q1s/bfe/PS4 bqq/six)
Ensemble simple-avg	Defensive averaging (Sreeram V)	Uniform mean of 4 estimators' OOS predictions per row	Used when no single estimator dominates
Multi-task NN (S4)	NN family (rubric, Family B)	Embedding(11,8) → Linear(d+8,64)+ReLU+Dropout(0.1)	On-gd linear (64,32) → Linear(13,06) shared → 11 per-ins

Champion architecture (plan §4.3 + Harry §4.11): per instrument, the champion is the best (pool, model) combination chosen by CPCV-mean AUC with the 1-SE rule (most-regularised within 1 SE preferred). Pools per Harry's INSTRUMENT_REGIMES: individual or asset-class pool, plus sub-class pools (energy_cl_ho, precious) for relevant instruments.

Cross-validation throughout: CombinatorialPurgedCV(n_groups=6, n_test_groups=2) → 15 paths. Per-instrument embargo via results/sreeram_experimental/instrument_scope.json. Modelling sample = events with t_signal ≤ 2021-10-06. Sealed test = t_signal > 2021-10-20 (10-day embargo). NO data after embargo touches model selection.

5. Cluster importance (S5 — 4 bug fixes implemented)

Method: Hygiene drop (NaN > 30%, zero-variance, $|p| > 0.99$ twin) → **Mantegna distance** $\sqrt{(1-|p|)}$ (bug fix 4) → Ward linkage + silhouette K → RF with **max_features='sqrt'** (bug fix 1) inside CPCV(6,2) (**bug fix 2:** purging) → Cluster MDI + clustered **purged MDA** (joint permutation per cluster) + **TreeSHAP via XGBoost native pred_contribs** (bug fix 3 — no shap library needed). Cross-method Kendall τ .

Class	Hygiene	K (sil.)	Top cluster MDA	n cluster > 0.02 MDA	alken comparable	Gate
Equity	70 / 80 kept	16	0.0223	1 / 16	0.025 (alken)	PASS
Energy	73 / 80 kept	8	0.0364	1 / 8	0.003 (alken)	PASS (12× alken)
Metals	71 / 80 kept	16	0.0155	0 / 16	-0.011 (alken)	CHECK (we beat alken)

Energy's dominant cluster (MDA 0.036, 19 features): *f19_atm_iv_1m, f19_atm_iv_3m, f19_iv_term_slope, f19_iv_pctile_252, f2_vol_20/60, f2_parkinson_20, f2_garman_klass_20, f2_vol_of_vol_20, f10_hl_range, f12_hurst_100, f9_dispersion_z_60, f9_pair_corr_mean_63, f9_implied_corr_z_252, f21_crack_321, f11_eia_crude_stock_rank63, f11_curve_slope, f11_vix_term_slope, f16_regime_alignment_score*. The F19 BBG options-IV pull (which alken did NOT have) is the empirical driver of energy outperformance.

Cross-method agreement (Kendall τ): SHAP ↔ MDI strong (0.72, $p < 1e-4$ on metals); SHAP ↔ MDA weak (0.15). Expected pattern per alken §5.16: MDI/SHAP are in-sample attribution, MDA is the OOS reality check.

6. Performance matrix — per-class CPCV mean AUC across methodologies

Methodology	Equity	Energy	Metals	Alken Δ	Notes
Alken (published)	0.579	0.525	0.530	—	Reference baseline
Per-class XGB only (S3.0)	0.521	0.556	0.512	-0.058 / +0.031 / -0.018	No instrument one-hot
+ instrument 1-hot + LGBM + ensemble (S3.1)	0.554	0.558	0.525	-0.025 / +0.033 / -0.005	Roster of 4 + simple-avg
+ champion arch. per inst. (S3-fix)	0.550	0.602	0.533	-0.029 / +0.077 / +0.003	Per Harry's INSTRUMENT_REGIMES; 1SE pick
+ multi-task NN (S4)	0.550	0.602	0.554	-0.029 / +0.077 / +0.024	gc1s rescued 0.48→0.57

Per-instrument champion AUC (with_bbg, sorted by AUC)

Inst	Champion pool	Champion model	AUC	± SEM	Lower 1SE CI	vs Harry post-refresh
cl1s	cl1s	LightGBM	0.671	0.022	0.649	Harry 0.675 (MATCH)
ng1s	energy_all	elastic-net log	0.601	0.044	0.557	Harry no signal
ho1s	energy_cl_ho	elastic-net log	0.599	0.048	0.551	Harry n/a (thin)
pl1s	pl1s	RandomForest	0.581	0.019	0.562	Harry 0.608 (-0.027)
gc1s	precious	multitask_nn	0.568	0.029	0.540	Harry 0.478
fesx1s	equity_all	RandomForest	0.557	0.019	0.538	Harry 0.579 (-0.022)
es1s	es1s	RandomForest	0.555	0.017	0.537	Harry 0.516
rb1s	rb1s	RandomForest	0.538	0.020	0.517	Harry 0.551
nq1s	equity_all	RandomForest	0.537	0.011	0.526	Harry 0.689 (-0.152) †
si1s	si1s	elastic-net log	0.534	0.011	0.522	Harry 0.515
hg1s	metals_all	RandomForest	0.533	0.014	0.519	Harry 0.604 (-0.071) †

† Harry's lead on nq1s/hg1s correlates with his wider GARCH labels (pt=1.5,sl=1.0); our spec is tighter per plan §3.2. Switching label spec is a §3.2 deviation; see decisions.

Acceptance gates

Gate	Target	Actual	Status
≥6 of 11 per-inst AUC > 0.55	≥ 6	7 (cl/ng/ho/pl/gc/fesx/es)	PASS
Signal flag (lower 1-SE CI > 0.50)	≥ 6	11 / 11	PASS
Per-class AUC ≥ alken + 0.02 (equity)	≥ 0.599	0.550	CHECK
Per-class AUC ≥ alken + 0.02 (energy)	≥ 0.545	0.602	PASS (+0.057)

Gate	Target	Actual	Status
Per-class AUC $\geq$ alken + 0.02 (metals)	≥ 0.550	0.554	PASS (+0.004)
R-11: BBG-missingness $\Delta \leq 0.03$	All classes	Max Δ 0.011	PASS
S5: ≥ 1 cluster > 0.02 MDA per class	All classes	2 of 3 (metals 0.016)	PASS / CHECK
S5: all 4 bug fixes present	All	All 4 in code + tests	PASS

S6 backtest (sealed test slice, 179 days)

Metric	Value	Notes
n days	179	Sealed test window (post-2021-10-20 embargo)
Ann. return (net)	17.3 %	After Grinold-Kahn costs (2 bps half-spread + 10 bps impact)
Ann. vol	5.40 %	Below 10 % R7 cap; slightly below [0.06, 0.10] target band
Sharpe	3.20	Per-period; pre-deflation
Sortino (full-T, Sortino-Price)	6.03	NOT std-of-negatives mis-impl
Max DD	-2.33 %	Shallow
Turnover (1/yr)	16.6	Holding ~6.6 days mean
Total cost (period bp)	142 bp	Gross 19.3 % $\rightarrow$ Net 17.3 %

Pre-deflation Sharpe 3.20 reads 'too good'. Plan §13 R-8 + alken's five-lens framework (AUC, MDA, significance, deflation, PT) is the discipline. S7 (significance + deflation + PT) is the next stage where we test if 3.20 survives the studentised block-bootstrap CI / DSR ladder / CSCV-PBO / Pesaran-Timmermann checks. Until then the 3.20 is one-lens and not a deployable claim.

7. Key decisions (bullet form)

Data + labels

- **Triple-barrier with t+1 entry** (Harry's load-bearing fix) — $\text{corr}(s_t, r_{t+1}) > 0$ on all 11 instruments empirically. Our 4886 events EXACT match Harry's events.csv per instrument.
- **GARCH(1,1) one-step daily σ** (refit 21d, min_obs=500, max_window=2000) — forward-aware, NOT EWMA (alken's default). Per-bar daily; barrier widths = $pt \cdot \sigma \cdot \sqrt{h}$.
- **pt=sl=0.5, h=10** — symmetric tight (NOT alken pt=sl=1.0 or Harry pt=1.5/sl=1.0). Resolves 87.5 % of events at PT or SL (vs alken ~50 % vertical). Plan §3.2 spec.
- **Per-instrument concurrency on each instrument's own trading-day index** — AFML Ch.4 uniqueness weights; calendar gaps don't inflate overlap.
- **Right-edge truncation: drop events whose [t+1, t+1+h] window isn't fully observable** — matches Harry's convention (4886 vs 4975 with truncation kept).
- **Harry-spec ablation tried, NOT shipped:** wider barriers lift ng1s to 0.800 but per R-10 that's construction-artefact (11 % label flip vs raw BBG). Artefacts retained for inspection only.

Features

- **F11 macro REBUILT:** 63-day rolling RANKS instead of raw LEVELS (alken's F11 had median KS 0.36, max 1.0 — catastrophic drift). Plus chg5/chg20 changes + 3 spread features.
- **F18 / F19 / F22 NEW** (Bloomberg-derived): term-structure, options-implied vol, EIA event flag. F19 options IV explicitly drives energy's outperformance (~6.6 % importance, in the highest-MDA cluster).
- **F21 cross-asset RV** (NEW, OHLCV-derived): gold/silver, copper/gold, crack 3-2-1. No Bloomberg needed.
- **Drift filter** (plan §3.3): DROP if $KS > 0.25$ AND $val_AUC < 0.54$, OR if $KS > 0.20$ AND $val_AUC < 0.51$. Brings 105 registered → 80 kept; 66 of 80 have $KS < 0.20$ (82.5 %, S2 gate PASS).
- **F17 HMM** (Sreeram): implementation failed (hmmlearn convergence) — emits NaN, drift-filtered out. Online EWMA HMM (alken's regime.py port) used instead — causal, no CV-seam.
- **R-11 ablation:** shipped *with_bbg*; without_bbg AUC delta ≤ 0.011 across classes (model is BBG-robust to H2-2022 missingness at hidden-test time).

Models + selection

- **Four-estimator roster:** elastic-net logistic, XGBoost, LightGBM, RandomForest — rubric requires linear + tree + NN families; the multi-task NN serves the NN slot.
- **Champion architecture per Harry §4.11 INSTRUMENT_REGIMES:** each instrument has 1-3 candidate pools (individual / sub-class / asset-class). The full 4xpool grid per instrument is searched.
- **1-SE selection rule** (plan §4.19): most-regularised model within 1 SE of best mean AUC is the champion. Rank: logistic < RF < LightGBM < XGB < multitask. Pool size: individual < sub-class < class.
- **Inner-CV XGB tuning** (plan §3.1 + §4.19): purged inner k-fold (k=4) over a 5-config grid ($\text{max_depth} \times \text{n_estimators} \times \text{reg_}\alpha / \text{reg_}\lambda$); 1-SE pick most regularised within 1 SE.
- **Instrument one-hot dummies** in the asset-class pool models (plan §3.1) — lets the tree learn instrument-conditioned splits without per-instrument training.
- **Multi-task NN** (plan §3.1 Family B): only candidate for multi-instrument pools (sharing meaningless for individual). Won gc1s precious champion (0.48 → 0.57).
- **Prediction clip [0.01, 0.99]** — bounds log-loss; AUC unchanged (monotone). Fixed energy logistic's blown-up log-loss (2-5 → ≤ 1.5).
- **CombinatorialPurgedCV(6, 2) → 15 paths** with per-instrument embargo via embargo_p90 map. Nested CPCV implemented + unit-tested but real-data run deferred (alken parity).

Cluster importance (S5)

- **Bug fix 1: max_features='sqrt'** — PS4's 'auto' was removed in sklearn ≥ 1.3 .
- **Bug fix 2: PurgedKFold for MDA** — KFold(shuffle=True) leaks across overlapping triple-barrier labels.
- **Bug fix 3: TreeSHAP via XGBoost native pred_contribs** — runs Tree SHAP internally; NO shap library / numba / llvmlite dependency required. The chain was blocked: shap requires numba which requires numpy < 2.4, but pandas 3.0 pins numpy ≥ 2.4 . XGBoost's built-in path sidesteps it.
- **Bug fix 4: Mantegna distance $\sqrt{(1-|p|)}$** — proper metric (Mantegna 1999); the original $1-|p|$ is non-metric.

- **Per-class K selection by silhouette** over [3, 16] — equity 16, energy 8, metals 16.
- **Energy top cluster MDA 0.036** (19 features, includes ALL F19 BBG IV) vs alken 0.003 — 12× alken. **This is where the BBG pull pays off.**
- **Cross-method Kendall τ :** SHAP $\leftrightarrow$ MDI = 0.72 (strong, both in-sample); SHAP $\leftrightarrow$ MDA = 0.15 (weak, MDA is OOS reality check) — matches alken §5.16's framing.

Risks documented in plan §13

- **R-1** BBG pull thin / unavailable: F18/F19/F22 silently drop. **HANDLED** — pulled and kept.
- **R-8** Grinold ceiling: pooled IC $\approx 0.07 \rightarrow$ AUC ceiling ≈ 0.52 . **Per-class AUC 0.55-0.60 is at the ceiling.**
- **R-9** BBG vs OHLCV convention: confirmed in practice — OHLCV is ratio back-adjusted; F18 uses BBG raw front + BBG raw 2nd, never mixed with OHLCV.
- **R-10 (NEW) Continuous-contract artefact on target**, esp. ng1s (R^2 0.72 vs raw, 11 % label flip). **Methodology must NOT claim raw-market profitability.** ng1s flagged in coverage_caveat.csv.
- **R-11 (NEW) H2-2022 BBG missingness:** simulated-missingness ablation built into pipeline. $\Delta \leq 0.011$ across classes $\rightarrow$ model is robust. Ship with_bbg. **Pulling H2-2022 BBG would VIOLATE the brief (§5.1).**
- **R-12 (deferred $\rightarrow$ resolved)** TreeSHAP bug fix 3 — resolved via XGBoost native pred_contribs, no external shap dep needed.

Where the artefacts live

- **S1 events:** data/sreeram_experimental_events.parquet (4886 rows $\times$ 10 cols)
- **S2 features:** data/sreeram_experimental_features.parquet (4886 $\times$ 80 + schema)
- **S2 drift audit:** results/sreeram_experimental/feature_drift_audit.csv
- **S3 instrument scope:** results/sreeram_experimental/instrument_scope.json
- **S3 champions (with_bbg):** results/sreeram_experimental/champions_summary.csv + champions_per_pool_per_model.csv
- **S3 R-11 ablation:** results/sreeram_experimental/bbg_missingness_ablation.csv
- **S3 coverage caveats:** outputs/coverage_caveat.csv (ng1s low_coherence_vs_raw flag)
- **S5 cluster importance per class:** results/sreeram_experimental/importance/{equity,energy,metals}/clustered_importance.csv + cluster_membership.csv + rank_agreement.csv
- **S6 deliverables:** outputs/metamodel_predictions.csv (calibrated), outputs/metamodel_predictions_raw.csv, outputs/strategy_weights.csv, outputs/experiment_log.csv
- **S6 backtest metrics:** results/sreeram_experimental/backtest_metrics.csv; daily returns: strategy_daily_net_returns.csv + strategy_daily_gross_returns.csv
- **Plan + tracker:** reports/sreeram_experimental/plan.md (golden record), action_tracker.md (PM-1..PM-4 log)
- **BBG validation:** reports/sreeram_experimental/bloomberg_validation_report.md, bloomberg_pull_list.md

Outstanding / not yet built

- **S7 significance + deflation + PT** (the other 4 lenses of alken's framework): modules *significance.py*, *deflation.py*, *signal_analysis.py* are **written** (studentised stationary block bootstrap, DSR ladder, CSCV-PBO with C(16,8)=12870, Pesaran-Timmermann, Treynor-Mazuy, Henriksson-Merton proxy) but the end-to-end significance_summary.md run is pending. Backtest Sharpe 3.20 needs to survive these gates.
- **S8 final academic report** with Harvard refs (alken's T3_03 equivalent): pending.
- **TorchVSN** (alken §5.12): not built. Multi-task NN already provides the per-instrument NN family slot.

Build state: 22 commits on Sreeram_experimental, all pushed to origin. 97 fast tests + 3 slow tests pass. Working tree clean.